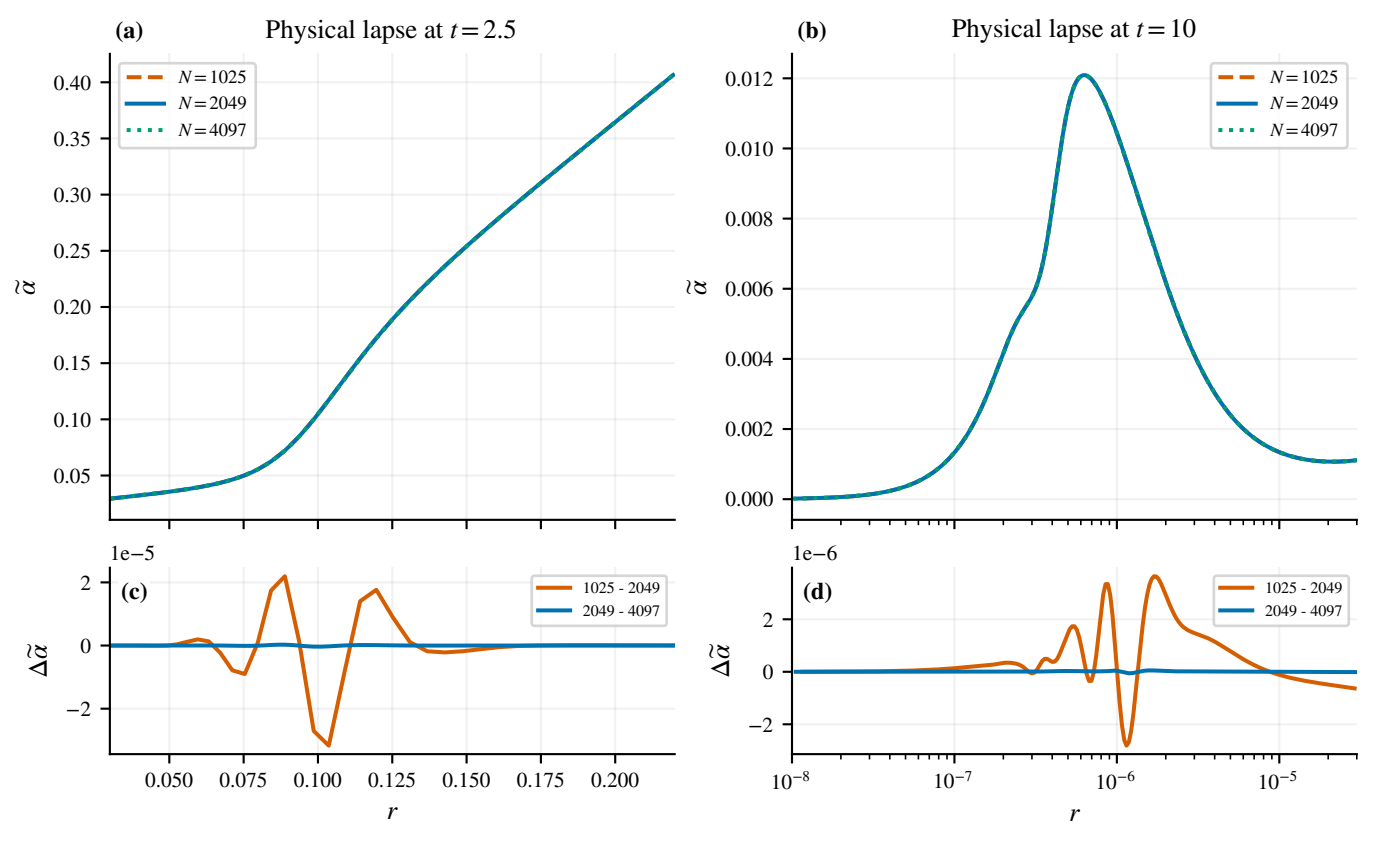
(a)
Physical lapse at $t = 2.5$
$N = 1025$
$N = 2049$
$N = 4097$
$\tilde{\alpha}$
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
1e−5
(c)
1025 - 2049
2049 - 4097
$\Delta\tilde{\alpha}$
2
0
−2
0.050
0.075
0.100
0.125
0.150
0.175
0.200
r
(b)
Physical lapse at $t = 10$
$N = 1025$
$N = 2049$
$N = 4097$
$\tilde{\alpha}$
0.012
0.010
0.008
0.006
0.004
0.002
0.000
1e−6
(d)
1025 - 2049
2049 - 4097
$\Delta\tilde{\alpha}$
2
0
−2
10^{-8}
10^{-7}
10^{-6}
10^{-5}
r